

Meyyappan Muthukaruppan

Lead Software Engineer / Staff Engineer

Lead / Staff Full-Stack Software Engineer with **12+ years** of industry experience delivering distributed, event-driven platform systems using scalable .NET microservices and Angular-based applications. Hands-on engineer with experience building AI agents using RAG-based architectures, LangChain, and MCP tools to extend platform capabilities. Provides technical leadership through platform direction, roadmap execution, cross-functional collaboration, and team mentorship.

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India (Relocating to Regional NSW)

491 Visa - Full Rights

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PROFESSIONAL EXPERIENCE

Trimble Inc.

Staff Software Engineer

Chennai, India
Aug 2018 – Present

Platform Engineering & Cloud-Native Systems Development

- Built and evolved **distributed, event-driven** common platform services using **.NET microservices, MongoDB, and Angular**, deployed across **AWS and Azure** to support multiple Forestry product lines.
- Contributed to **greenfield platform architecture and modernisation**, defining service boundaries, API contracts, data ownership, and deployment models aligned with platform strategy.
- Modernised a legacy monolithic desktop application into a **multi-tenant, cloud-ready SaaS platform** using a strangler pattern and **domain-aligned microservices**.
- Remained hands-on, designing and implementing **RESTful APIs**, backend services, and **Angular front-end architectures** with shared UI modules and reusable components.
- Designed deployment strategies for **multi-tenant SaaS and single-tenant customer-hosted** environments, addressing scalability, isolation, and compliance requirements.
- Implemented **event-driven and serverless workloads** using **AWS Lambda and Azure Functions**, and established **CI/CD pipelines and infrastructure provisioning** to enable reliable, repeatable releases.

AI Enablement & Intelligent Platform Capabilities

- Built **AI agents** integrated with existing platform services using **Retrieval-Augmented Generation (RAG)** and **LLM-based workflows** to enable context-aware automation.
- Implemented **LangChain** for agent orchestration, including tool execution, prompt chaining, and controlled interaction with platform APIs.
- Designed **MCP-style tools** exposing existing application endpoints as callable interfaces for AI agents, enabling safe, governed automation.
- Implemented **document-based RAG pipelines** using platform documentation (user manuals, configuration and installation guides) with **embeddings and vector search**.
- Led the **adoption of Cursor AI** across the engineering team, defining usage rules, security guardrails, and onboarding practices for responsible AI-assisted development.

Technical Leadership, Strategy & Delivery

- Influenced platform technical direction through **architecture reviews, design discussions, and modernisation initiatives**, ensuring consistent adoption of common platform services.
- Partnered with **Product Management and UX** to translate business requirements into scalable, technically feasible platform solutions.
- Supported **roadmap planning and delivery execution**, improving sprint predictability, release stability, and alignment across dependent teams.
- Embedded **quality, security, and reliability practices** into delivery workflows, supporting consistent and stable releases.
- Mentored engineers across levels, providing technical guidance, design feedback, and supporting professional growth.
- Promoted **cross-team collaboration and innovation** through technical forums, knowledge sharing, and Forestry-level hackathons.

Conduent (formerly Xerox)

Senior Software Engineer

Chennai, India

Nov 2016 – Aug 2018

- Contributed to the development and enhancement of **Agile Court 8.0**, a mission-critical **judicial case management system** used by **50+ US State and County courts** to manage end-to-end case lifecycle workflows.
- Participated in the **migration from .NET Framework to .NET Core**, modernising backend components into **RESTful APIs** to improve scalability, maintainability, and deployment flexibility.
- Designed and implemented **Angular-based UI modules** and backend services using **.NET Core, SQL Server**, and **OAuth/claims-based authentication**, translating complex legal and operational requirements into secure, maintainable application components.
- Optimised application performance for **large datasets (100,000+ active cases)**, improving query efficiency, response times, and overall system responsiveness.
- Worked within **Agile/Scrum** teams, collaborating with business analysts, product owners, QA, and external vendor teams to deliver features on schedule and support cross-system interoperability.
- Supported **code quality and security practices** through peer reviews, test automation, and adherence to secure coding standards suitable for **government-regulated systems**.

Span Technology Services

Senior Software Engineer

Chennai, India

May 2013 – Nov 2016

- Designed and built an **IRS e-filing engine** supporting US federal tax compliance, converting customer data into **IRS-mandated formats** (MeF, FIRE, AIR) for filings such as **1099, W-2, W-9, 1095, and 94x**, used by **10,000+ business clients**.
- Developed a **multi-tenant web application** with well-defined database schemas, business logic layers, and **RESTful APIs**, ensuring data accuracy, audit compliance, and scalability during **peak tax seasons (500,000+ annual submissions)**.
- Implemented **secure data transmission and encryption** for IRS web service integrations, meeting federal security and audit requirements with consistent compliance in production.

- Built **RESTful backend services** used by **iOS and Android applications**, enabling reliable data synchronisation, shared business logic, and offline-friendly user workflows.
- Developed **Application-to-Application (A2A)** transmission services using **WSE-based architectures** to support secure, automated system-to-system IRS filings.
- Led the design and implementation of an **e-filing interview workflow engine**, handling complex tax rules, validations, and conditional logic to ensure accurate and error-free submissions.

TECHNICAL SKILLS

- **Languages & Frameworks:** .NET (C#), JavaScript / TypeScript, Python, Angular
- **AI & Intelligent Systems:** RAG, LangChain, LLM integration, vector databases, embeddings, prompt engineering, AI agents, MCP-style tools
- **Data & Databases:** MongoDB, SQL Server
- **Cloud Platforms:** AWS (Lambda, EC2, S3), Azure (Functions, App Services, Service Bus, WebHooks)
- **Architecture & Design:** Microservices, RESTful APIs, multi-tenant SaaS, cloud-native, serverless systems
- **DevOps & Security:** CI/CD, Git, Docker, Infrastructure as Code, DevSecOps, SAST / SCA
- **Domain & Platform Capabilities:** ESRI ArcGIS, Power BI integrations, Single Sign-On (SSO), IRS e-filing systems
- **Delivery & Leadership:** Agile / Scrum, platform engineering, technical leadership

VISA, WORK RIGHTS & LOCATION

- **Skilled Work Regional (Provisional) Visa (Subclass 491 – NSW Nomination)**
- Full work rights in Australia
- Eligible to live and work in **NSW regional areas**
- Open to **hybrid or on-site** roles aligned with visa conditions

EDUCATION

- **Bachelor of Engineering (Electronics & Communication Engineering)**
Anna University, India | 2008 – 2012